MedAgent Clinical Decision Support

Case ID: test_case_int | **Status:** APPROVE | **Physician:** Dr. House | **Date:** 2026-10-16 06:53 UTC

CRITICAL DISCLAIMER: Decision-support draft generated by AI agents. It is not a diagnosis and must be reviewed, corrected and approved by a licensed physician before any clinical use.

Patient Intake & Presentation

Patient: Age: 52, Sex: Female
Notes & History: Fever, productive cough, crackles.

Clinical Summary / Impression

Clinical evaluation indicates features consistent with Community-Acquired Pneumonia, characterized pathophysiologically by acute microbial infection triggering inflammatory exudation and leukocyte accumulation in pulmonary alveolar spaces, causing parenchymal consolidation and impaired oxygen diffusion. Key supporting clinical indicators include fever, cough, crackles, while secondary differential considerations include Acute Viral Bronchitis / Atypical Infection. Targeted diagnostic workup and guided serial clinical monitoring are recommended to establish definitive management.

Ranked Differential Diagnoses

#	Condition	Likelihood	Key Supporting Findings	Citations
1	**Community-Acquired Pneumonia**	**HIGH**	fever, cough, crackles, Leukocytosis, Leukocytosis	S1, S2
2	**Acute Viral Bronchitis / Atypical Infection**	**LOW**	Respiratory presentation, Cough and dyspnea	S3

Laboratory Findings & Identified Patterns

Analyte / Test	Result	Unit	Reference Interval	Flag
WBC Count	**14.5**	x10^3/uL	4.5 – 11	**HIGH**

Inferred Diagnoses (ClinicalLabNet Neural Classifier):
• **[MODERATE] Type 2 Diabetes Mellitus / Severe Hyperglycemia** [60.0% Neural Confidence] (Likelihood: LOW): Neural Network Confidence: 60.0% (ClinicalLabNet deep multi-label classifier). Biomarker pattern is characteristic of type 2 diabetes mellitus / severe hyperglycemia. *Pathophysiology:* Chronic metabolic disorder of peripheral insulin resistance and progressive pancreatic beta-cell secretory failure resulting in sustained hyperglycemia.

Synthesized Clinical Patterns:
• **[ROUTINE] Leukocytosis:** Elevated white blood cell count indicating active inflammatory, infectious, or reactive marrow response.

Suggested Diagnostic Workup

• Sputum gram stain and culture
• Blood cultures prior to antibiotics
• Follow-up chest radiograph in 6 weeks
• Respiratory viral multiplex PCR

Physician Attestation: I, Dr. House, hereby confirm that I have reviewed, verified, and adjusted this decision-support synthesis. Decision: **APPROVE**.
Reviewer Notes: Clinical correlation confirms bacterial pneumonia.
Signature: _____ **Date:** 2026-10-16 06:53 UTC